Ekpyrotic Contraction, Anisotropy Smoothing, and WKB Classicality: a Controlled Numerical Study

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Abstract

We reproduce, with quantified numerical error control, the classical dynamics of an ekpyrotic contracting universe sourced by a canonical scalar field in a negative exponential potential, $V(\phi) = -V_0 e^{-c\phi}$, $c = \sqrt{2/p}$. Integrating the flat Friedmann–Raychaudhuri system in contraction e-folds $n = -\ln a$, with anisotropy modelled as an adiabatic shear energy density $\rho_\Sigma \propto a^{-6}$, we verify the smoothing law $\rho_\Sigma/\rho_\phi \propto a^{3(w-1)}$ and $\rho_\phi \propto a^{-3(1+w)}$ with fitted exponents matching the analytic attractor predictions to 4–6 significant digits across $w \in \{65.7, 12.3, 5.7, 1.2\}$ ($p = 0.01 \ldots 0.3$), and confirm with a $w = 1/3$ counterexample that $w > 1$—not contraction per se—is the operative condition. A change of variables to t-invariant relative coordinates reduces the flow about the scaling solution to an autonomous system with the exact spectrum $\lambda = \{0, -\frac{3}{2}(w-1), -3(w+1)\}$: deviations decay as $a^{3(w-1)/2}$ (half the smoothing exponent), the zero mode is the time-translation direction, and $w = 1$ emerges as the *linear stability boundary* of the scaling solution—the same threshold that governs smoothing. We then isolate the additional assumptions needed for a bounce: a separately specified effective Friedmann equation $H^2 = \frac{\rho}{3}(1-\rho/\rho_c)$ of loop-quantum-cosmology type bounces exactly at $\rho = \rho_c$, but with $V < 0$ and no further physics it recollapses almost immediately ($a_{\max}/a_b = 1.0115$), and its continuation is a Zeno cascade whose structure we derive exactly: cycle durations $\Delta t_n = A/n^2$ with $A = 3p/\pi\sqrt{3\rho_c}$, driven by a constant velocity growth $\alpha = c\pi\sqrt{\rho_c/3}$ per cycle—both confirmed to four significant digits—so infinitely many bounces with unbounded kinetic energy accumulate in finite time. Flipping the sign of the potential in the expanding branch does not rescue it: the post-bounce velocity points toward larger $|V|$ and the solution is driven back across the constraint boundary in finite time. Finally, we evaluate the Wheeler–DeWitt WKB classicality quantity $\mathcal{Q} = \hbar|\nabla^2 S|/|(\nabla S)^2|$ along the classical ekpyrotic trajectory and verify its attractor scaling $\mathcal{Q} \propto a^{(1-3p)/p}$: the semiclassical description of the contracting branch improves monotonically through ekpyrosis for $p < 1/3$. We also document a numerical-reliability finding: the attractor's marginal time-translation direction converts any homogeneous perturbation into a crunch-time displacement $\delta t_c \propto f$, linear over three decades, leaving t-invariant observables unaffected. All results are reproducible from the accompanying code and executed notebook.

1 Introduction

The ekpyrotic scenario [1] proposes that the hot big bang emerged from a phase of slow contraction driven by a scalar field with an ultra-stiff effective equation of state, $w \gg 1$. Its central classical claim is a smoothing mechanism that is, in a precise sense, dual to inflation: since the scalar-field energy density scales as $\rho_\phi \propto a^{-3(1+w)}$ while anisotropy, curvature, and radiation scale as a^{-6}, a^{-2}, and a^{-4} respectively, the ratio of any such energy component to the scalar density behaves as a nonnegative power of a during contraction when $w > 1$:

$$\frac{\rho_\Sigma}{\rho_\phi} \propto a^{3(w-1)} \quad \Longrightarrow \quad \text{shrinks during contraction if } w > 1. \tag{1}$$

Contraction therefore approaches an isotropic, homogeneous, scalar-dominated attractor [2], whereas with $w < 1$ a contracting universe is shear dominated (a^{-2} growth of the ratio for free shear—the mixmaster direction).

The scenario's open classical and semiclassical questions are equally well known. (i) General relativity offers no exit from contraction: classically the density diverges at finite time, and every concrete nonsingular completion (ghost condensation, galileons, string-inspired modifications, quantum-gravity effective dynamics) adds assumptions on top of the GR model [3, 5]. (ii) Whether the quantum description of the contracting phase is *classical* in the sense required for a cosmological background—peaked on classical histories—has been analysed through WKB criteria applied to the Wheeler–DeWitt (WDW) equation; for exponential potentials the classicality factor was found to behave as $e^{-(\epsilon-3)\mathcal{N}/(\epsilon-1)}$ with fast-roll parameter ϵ, so that ekpyrosis ($\epsilon > 3$) classicalises exponentially in the number $\mathcal{N}$ of contraction e-folds [6].

This paper is a controlled numerical study of these three layers of the problem, with the aim of separating what the simplified classical model actually predicts from what additional assumptions buy:

1. reproduce the contraction and the smoothing law (1) for several equations of state, including a deliberately included $w < 1$ counterexample;

2. quantify anisotropy evolution and the accuracy of the integration (constraint residuals, tolerance and parameter sweeps);

3. contrast the GR contracting solution—which crunches—with a separately specified effective bounce model ($H^2 = \frac{\rho}{3}(1 - \rho/\rho_c)$, in the style of loop quantum cosmology [9]), and document what that model does not deliver;

4. evaluate the WKB classicality quantity $\mathcal{Q} = \hbar|\nabla^2 S|/|(\nabla S)^2|$ along the numerical ekpyrotic trajectory and compare with the attractor prediction, connecting the local (trajectory-level) aspect of the classicality result of Ref. [6];

5. document assumptions, predictions, and what remains unverified.

Two of our findings go beyond reproduction. First, the specified effective bounce model, taken at face value with a negative potential, *fails in an instructive way*: it recollapses almost immediately after the bounce and, if continued, enters a Zeno cascade of bounces whose structure we derive exactly—cycle durations $\Delta t_n = A/n^2$ with $A = 3p/\pi\sqrt{3\rho_c}$, driven by a constant velocity growth $\alpha = c\pi\sqrt{\rho_c/3}$ per cycle—accumulating in finite time with unbounded kinetic energy: visible evidence of where the ad-hoc construction stops being a model at all. Second, a change of variables to t-invariant relative coordinates reduces the flow about the scaling solution to a three-dimensional autonomous system whose exact spectrum we obtain: deviations decay as $a^{3(w-1)/2}$ (half the smoothing exponent), the zero mode is the time-translation direction along which *any* homogeneous perturbation (including the small backreaction of shear at fraction f) produces a crunch-time displacement $\delta t_c \propto f$, measured linear over three decades ($\delta t_c/|t_i| \approx -36.8\, f$ for $p = 0.1$)—and the stability boundary of the spectrum sits exactly at $w = 1$, the same threshold that governs smoothing. The time-translation effect is a coordinate artefact with no physical content, but it silently corrupts any comparison of solutions in absolute cosmic time, and it dictates the integration variables we use.

All computations are reproducible: the model module, the notebook that produces every number and figure of this paper, and the machine-readable metrics are distributed with this work [10]. Units are reduced Planck units, $M_{\rm Pl} = 1$, so the Friedmann constraint reads $3H^2 = \rho$.

2 Model and method

2.1 Background and matter content

We consider a flat FLRW background, $\mathrm{d}s^2 = -\mathrm{d}t^2 + a^2(t)\mathrm{d}\mathbf{x}^2$, sourced by a canonical scalar field in the exponential potential

$$V(\phi) = s\, V_0\, e^{-c\phi}, \qquad c = \sqrt{2/p}, \tag{2}$$

with $s = -1$ in the ekpyrotic regime $p < 1/3$ and $s = +1$ for $p > 1/3$. The sign is forced: the exact power-law scaling solution

$$a \propto |t|^p, \qquad -c\phi = 2\ln\frac{|t|}{|t_i|} - c\phi_i, \qquad w = -1 + \frac{2}{3p}, \tag{3}$$

requires $V = p(3p-1)/t^2$, which has the sign of $3p-1$. The $V > 0$, $p > 1/3$ branch ($w < 1$) is included solely as a counterexample to isolate the role of the condition $w > 1$.

Anisotropy is modelled as a homogeneous shear energy density ρ_Σ. For a Bianchi I universe with isotropic source this is exact: the shear propagates as free gravitational radiation and

$$\dot{\rho_\Sigma} = -6H\rho_\Sigma, \qquad \rho_\Sigma \propto a^{-6}. \tag{4}$$

The Friedmann constraint and Raychaudhuri equation then read

$$3H^2 = \underbrace{\tfrac{1}{2}\dot\phi^2 + V(\phi)}_{\rho_\phi} + \rho_\Sigma, \qquad \dot H = -\frac{\dot\phi^2}{2} - \rho_\Sigma. \tag{5}$$

We evolve $\dot\phi = u$, $\dot u = -3Hu - V_{,\phi}$, (4) and the $\dot H$ equation of (5), with H integrated from Raychaudhuri so that the constraint residual $\mathcal{C} \equiv 3H^2 - \rho_{\rm tot}$ is an *independent* quantity monitoring numerical accuracy rather than a reimposed algebraic relation. Initial data are placed exactly on the scaling solution (3) at a large negative time t_i, normalised by $a(t_i) = 1$, with $\rho_\Sigma(t_i)/\rho_\phi(t_i) = f$ and H_i adjusted to the constraint surface, $3H_i^2 = \rho_{\rm tot}(t_i)$. Unless stated otherwise, $V_0 = 1$, $t_i = -10^6$, $f = 10^{-4}$.

2.2 Integration in contraction e-folds

Integrating in cosmic time t fails quietly: as $t \to t_c^-$ the time step collapses and the accumulated pointwise error pushes the solution onto a *time-shifted* copy of the attractor—exact in phase space, wrong in t (Sec. 5 explains why this is inevitable). We therefore integrate in contraction e-folds

$$n = -\ln\frac{a}{a_i} = -\ln a, \qquad \dot n = h \equiv -H > 0, \tag{6}$$

with state $y = (\phi, u, h, \rho_\Sigma, t)$ and

$$\frac{\mathrm{d}\phi}{\mathrm{d}n} = \frac{u}{h}, \quad \frac{\mathrm{d}u}{\mathrm{d}n} = 3u - \frac{V_{,\phi}}{h}, \quad \frac{\mathrm{d}h}{\mathrm{d}n} = \frac{u^2/2 + \rho_\Sigma}{h}, \quad \frac{\mathrm{d}\rho_\Sigma}{\mathrm{d}n} = 6\rho_\Sigma, \quad \frac{\mathrm{d}t}{\mathrm{d}n} = \frac{1}{h}. \tag{7}$$

The attractor (3) is exactly exponential in n (all components $\propto e^{n/p}$ or affine in n), so a single relative accuracy holds uniformly to arbitrary depth of contraction. Integration uses DOP853 (8th order, dense output) at relative tolerance 10^{-11}; the GR runs are stopped by a terminal event $\rho_{\rm tot} = \rho_{\rm stop} = 10^{10}$, which is a trust cutoff on the classical integration, not a feature of the solution.

Conversely, t itself is ill-conditioned along the attractor: the neutral time-translation mode of (3) amplifies errors in t as $e^{n/p}$ (Sec. 5). All physical statements in this paper are therefore made in t-invariant variables (n, a, ρ); t is used only where its accuracy is adequate (Secs. 3.3 and 5), with the reference crunch time measured from the trajectory as the late-time plateau of $t - p/H$.

2.3 Effective bounce dynamics

For the bounce comparison we adopt—as an explicitly *specified* dynamics, not a derivation—the effective Friedmann equation of the loop-quantum-cosmology phenomenology [9],

$$H^2 = \frac{\rho}{3}\Big(1 - \frac{\rho}{\rho_c}\Big), \tag{8}$$

with the continuity equation unchanged, and hence the unique consistent modified Raychaudhuri equation

$$\dot{H} = -\frac{\rho + p}{2}\Big(1 - \frac{2\rho}{\rho_c}\Big). \tag{9}$$

We integrate this system from the same attractor initial data, in cosmic time through the bounce (the dense region ranges over only ~ 2 e-folds, where cosmic-time integration is well conditioned, as confirmed by a tolerance sweep of the bounce location, stable to seven digits). The default critical density is $\rho_c = 0.41$, the standard effective value [9], with a ± 1 decade sensitivity sweep. The bounce is located by the event $H = 0$; the run is terminated at $\rho = 0$ if the expanding branch recollapses (it does). The system monitors its own modified constraint $\mathcal{C}_{\rm eff} = 3H^2 - \rho(1 - \rho/\rho_c)$, kept at $\sim 10^{-9}$ along the run.

2.4 WKB classicality

In minisuperspace ($\alpha = \ln a, \phi$), with the factor ordering chosen so that no first derivatives appear, the WDW equation reads

$$\Big[\hbar^2\Big(\frac{\partial^2}{\partial\alpha^2} - 6\frac{\partial^2}{\partial\phi^2}\Big) + 12\,a^6 V(\phi)\Big]\Psi = 0, \tag{10}$$

whose Hamilton–Jacobi equation is $-(\partial_\alpha S)^2 + 6(\partial_\phi S)^2 + 12a^6V = 0$. A WKB state $\Psi = Ae^{iS/\hbar}$ is classical when the transport (Vlasov) equation for the amplitude is phase dominated, quantified by [6, 7]

$$\mathcal{Q} \;\equiv\; \frac{\hbar\,|\nabla^2 S|}{|(\nabla S)^2|} \ll 1, \tag{11}$$

with $(\nabla S)^2 = -(S_{,\alpha})^2 + 6(S_{,\phi})^2 = -12a^6V$. Evaluating the classical momenta $S_{,\alpha} = -6a^3H$, $S_{,\phi} = a^3u$ and their derivatives along the *numerical* trajectory (using $\dot{u} = -3Hu - V_{,\phi}$; no attractor scalings enter the evaluation) gives $\mathcal{Q}$ pointwise. On the attractor, analytically,

$$\mathcal{Q} = \frac{\hbar}{2p}\frac{|t|}{a^3} \;\propto\; a^{(1-3p)/p}, \tag{12}$$

which for $p < 1/3$ falls to zero as the crunch is approached. With the fast-roll parameter $\epsilon = 3(1+w)/2 = 1/p$ this reads $\mathcal{Q} \propto e^{-(\epsilon-3)\mathcal{N}/(\epsilon-1)}$ in contraction e-folds $\mathcal{N} = \ln(a_i/a)$—precisely the exponential classicality factor of Ref. [6].

3 Results

3.1 Smoothing for $w > 1$; the $w < 1$ counterexample

Figure 1 and Table 1 summarise step 1. Fitted power-law exponents (log-log least squares in the window $\rho_\Sigma/\rho_\phi < 10^{-3}$) reproduce the attractor predictions $\rho_\phi \propto a^{-2/p}$ and $\rho_\Sigma/\rho_\phi \propto a^{3(w-1)}$ to 4–6 significant digits.

Two points deserve emphasis. First, the $p = 0.5$ run is a genuine counterexample, not a failed simulation: $\rho_\Sigma/\rho_\phi \propto a^{-2}$ initially, the shear grows relative to the field, backreacts, and eventually takes over the contraction entirely (Kasner-like domination). The condition

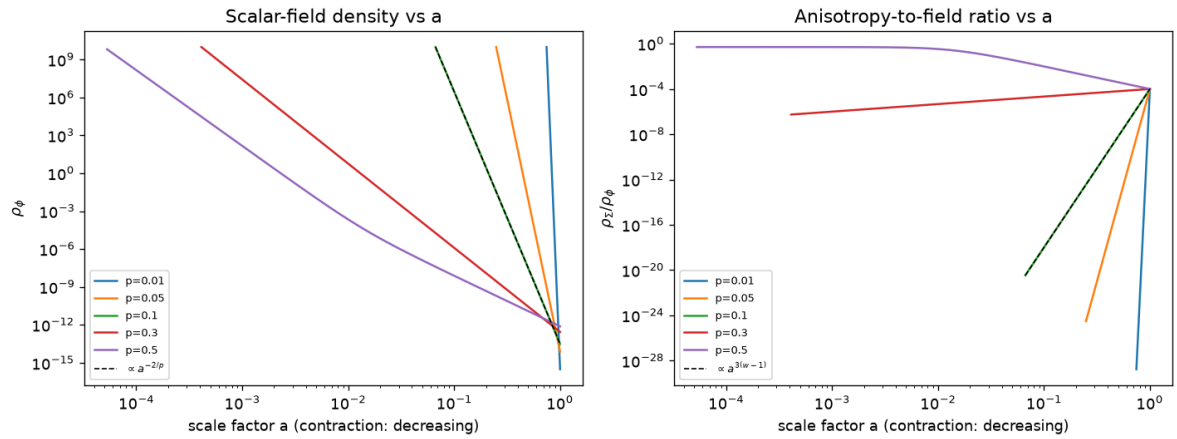


Figure 1: Contraction for different equations of state. Left: scalar-field density vs scale factor for $p \in \{0.01, 0.05, 0.1, 0.3, 0.5\}$ (initial data on the attractor, $a(t_i) = 1$; contraction proceeds to the left). Right: anisotropy ratio ρ_Σ/ρ_ϕ vs a. Dashed: analytic attractor slopes $a^{-2/p}$ and $a^{3(w-1)}$. The $p = 0.5$ curve ($w = 1/3$) shows the ratio *growing* as a^{-2} until the shear leaves the perturbative regime.

for smoothing is $w > 1$, not contraction. Second, the ekpyrotic phase is *short*: the end of the physically relevant contraction, at density ρ_f, occurs after $n = \frac{p}{2}\ln(\rho_f/\rho_i)$ e-folds—here only 1.51 e-folds to reach $\rho = 0.41$ for $p = 0.1$. Smoothing nonetheless succeeds with ease: from $f = 10^{-4}$ the ratio falls by a factor 6.4×10^{-10} over those 1.5 e-folds, precisely because the suppression per e-fold $e^{-3(w-1)}$ is enormous for $w \gg 1$. This exponential-in-$\mathcal{N}$ structure, shared with the classicality result of Sec. 3.4, is the operative fact, not the number of e-folds.

3.2 Anisotropy sensitivity and convergence

Figure 2 shows the tolerance sweep. The constraint residual scales with the tolerance as expected ($3.2 \times 10^{-6} \to 1.7 \times 10^{-12}$ relative to $\rho_{\rm tot}$ for rtol $10^{-7} \to 10^{-13}$), and the fitted smoothing exponent is stable at 13.99998 (theory 14) at all but the loosest tolerance.

Parameter sensitivity: varying the initial shear fraction over $f \in \{10^{-6}, 10^{-4}, 10^{-2}\}$ leaves the exponent unchanged (14.000000, 13.999978, 13.997786) and the final ratio scales linearly, $\{3.420, 3.420, 3.434\} \times 10^{-23,-21,-19}$: a 1% shear fraction at three decades below Planckian cutoff still leaves a sub-percent ratio at the cutoff. Varying V_0 over two decades changes nothing (13.999978 throughout), as expected since V_0 only redefines ϕ_i.

3.3 GR contraction versus a specified effective bounce

The GR solution crunches. The specified dynamics (8), (9), starting from identical initial data, follows it exactly until ρ approaches ρ_c and then bounces—cleanly, at $\rho = \rho_c$ to ten digits (Table 2). The bounce occurs at essentially the scale factor at which GR reaches the same

Table 1: Fitted vs analytic exponents. Fits in the perturbative window $\rho_\Sigma/\rho_\phi < 10^{-3}$; $R^2 \geq 0.99999999$; full data in `results/metrics.json`.

p	w	$\rho_\phi \propto a^{-3(1+w)}$: fit / theory	$\rho_\Sigma/\rho_\phi \propto a^{3(w-1)}$: fit / theory
0.01	65.67	-199.9999 / -200	193.9999 / 194
0.05	12.33	-40.0000 / -40	34.0000 / 34
0.10	5.67	-20.0000 / -20	14.0000 / 14
0.30	1.22	-6.6665 / -6.6667	0.6665 / 0.6667
0.50	0.33	-4.0007 / -4	-1.9993 / -2

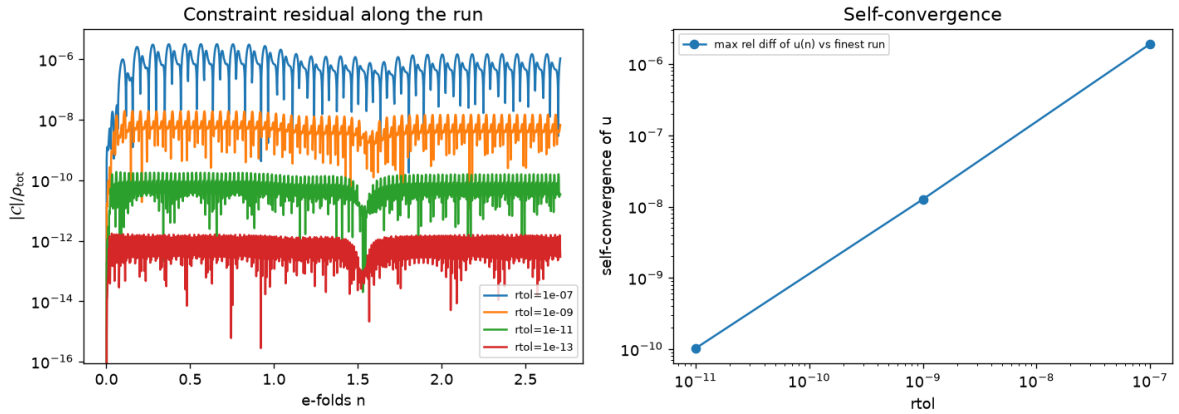


Figure 2: Numerical control. Left: independent Friedmann constraint residual $|\mathcal{C}|/\rho_{\rm tot}$ along the run ($p = 0.1$, in e-fold time) at four tolerance levels. Right: self-convergence of $u(n)$ against the finest run, showing ~ 2 digits gained per decade of relative tolerance.

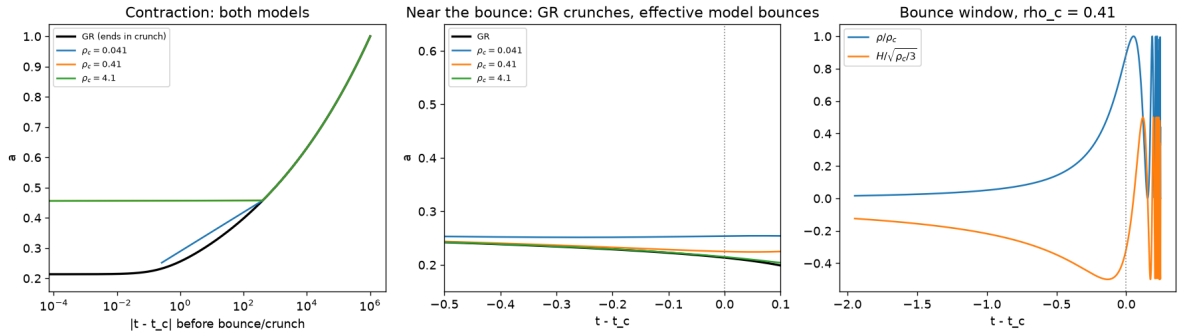


Figure 3: No bounce versus specified bounce. Left: a against time $|t - t_c|$ to crunch (GR) and to bounce (effective model, $\rho_c/0.41 \in \{0.1, 1, 10\}$); the curves are indistinguishable until $\rho \sim \rho_c$. Middle: the same solutions near the high-density region (crunch-centred time, see Sec. 5); the effective model bounces, the GR solution ends. Right: ρ/ρ_c and $H/\sqrt{\rho_c/3}$ for $\rho_c = 0.41$ through the bounce: the expanding branch recollapses ($\rho \to 0$) and, if continued, the model enters a cascade of bounces accumulating in finite time—the region of densely packed features at the right edge, which we treat as a breakdown of the ad-hoc dynamics.

density ($a_b/a_{\rm GR}$ within 2%; the difference is the modified source term acting over the densest stretch of the contraction).

The more informative result is what happens *after* the bounce. The potential being negative, the expanding branch cannot persist: the kinetic energy redshifts as a^{-6} and catches up with $|V|$ almost at once; the model recollapses after expanding by only

$$\frac{a_{\rm max}}{a_{\rm bounce}} \;=\; 1.0115 \qquad (\rho_c = 0.41,\; p = 0.1). \tag{13}$$

Continued further, the effective dynamics falls into a cascade of contraction–bounce–recollapse cycles whose period shrinks fast enough for an infinite number of bounces to accumulate at a finite time (Fig. 3, right). We classify the cascade as an artifact: it signposts the point where gluing (8) onto a negative exponential potential with unchanged continuity stops being a model. In realistic ekpyrotic constructions the bounce mechanism explicitly *changes* the dynamics in the high-energy phase (ghost condensation violates the null energy condition [3]; galileon bounces modify the kinetic sector [4]); none of that machinery is present here, and the cascade is the price.

The positive content of this comparison: (i) the bounce in the specified model lands exactly

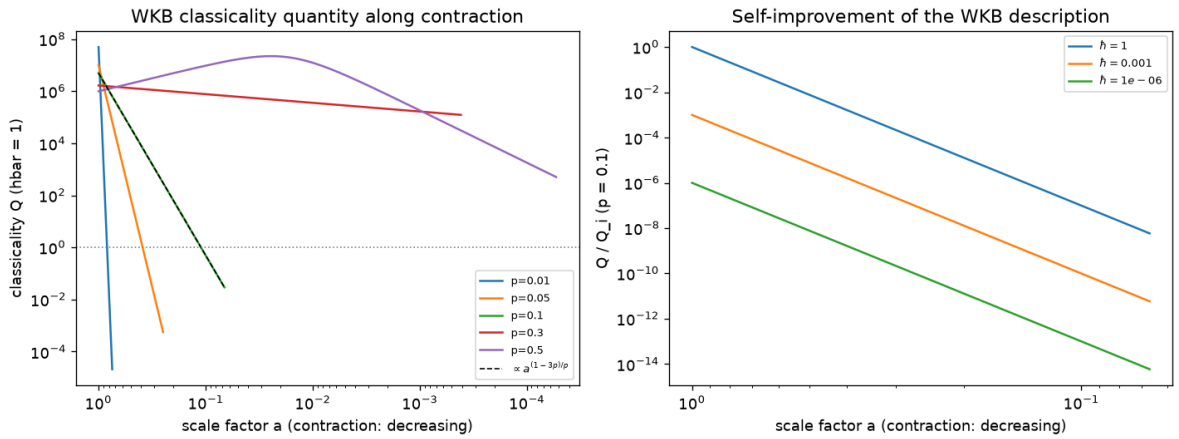


Figure 4: Classicality. Left: $\mathcal{Q}(a)$ along the contraction for the five equations of state ($\hbar = 1$); dashed, the attractor slope $a^{(1-3p)/p}$. For $p < 1/3$ the WKB description improves monotonically as the high-density region is approached; for $p = 0.5$ ($w < 1$) it degrades as the shear takes over. Right: same run ($p = 0.1$), normalised, for three values of the overall $\hbar$ factor: the exponent is $\hbar$ independent, the overall level is not.

where it should, which validates the implementation; (ii) smoothing is long finished before the bounce ($\sim 10^9$ suppression by $\rho = \rho_c$), so the post-bounce isotropy is inherited, not generated, by the bounce model; (iii) the failure mode is sharp and visible—which is the reason for including a model we know to be incomplete.

3.4 WKB classicality along the ekpyrotic branch

Figure 4 shows $\mathcal{Q}$ computed from the numerical trajectory via (11). The fitted power laws match the attractor prediction (12) to 4–6 digits for all p (Table 3). For ekpyrotic p, $\mathcal{Q}$ plummets: with $\hbar = 1$ in our units, $\mathcal{Q}(t_i) \sim 5 \times 10^6$ falls below 1 at $a/a_i \approx 0.11$ for $p = 0.1$. The $w < 1$ run degrades classicality ($\mathcal{Q} \propto a^{-1}$ in the perturbative window), consistent with the picture that what fails to smooth also fails to classicalise [6].

As derived in Sec. 2, our measured exponent is exactly the classicality factor $e^{-(\epsilon-3)\mathcal{N}/(\epsilon-1)}$ of Ref. [6]. Two caveats delimit what this establishes. First, $\mathcal{Q} \propto \hbar$ (it is the ratio of the transport to the Hamilton–Jacobi equation): the *exponent* is convention independent, the crossing point of $\mathcal{Q} = 1$ is not, and we quote $\hbar = 1$ explicitly. Second, (11) is a self-consistency criterion for a WKB state peaked on the given classical history in minisuperspace (first-derivative-free factor ordering, no fluctuations); it does not select among boundary proposals, nor address the quantum production of perturbations—the parts of the classicality debate carried, in Refs. [6, 7, 8], by saddle-point analyses of the wavefunction. What we verify here is the local dynamical input to those analyses: along the ekpyrotic history the WKB ansatz improves precisely as predicted, throughout the phase.

Table 2: Bounce locations ($p = 0.1$). $\rho(t_b) = \rho_c$ holds to ten digits in every case. $a_{\rm GR}(\rho_c)$ is the GR scale factor at the same density; the suppression column uses the GR scaling (1) up to that density.

ρ_c	$a_{\rm bounce}$	$a_{\rm GR}(\rho_c)$	ρ_Σ/ρ_ϕ suppression up to $\rho = \rho_c$
0.041	0.2513	0.2473	2.0×10^{-9}
0.41	0.2240	0.2204	6.4×10^{-10}
4.1	0.1996	0.1964	2.0×10^{-10}

4 Attractor stability, and why naive bounce continuations fail

4.1 Exact spectrum in t-invariant relative coordinates

The convergence of the contraction onto the scaling solution, and the marginal behaviour of Sec. 5, are both captured exactly by a change of variables. For the pure-scalar system ($\rho_\Sigma = 0$) define

$$X = -c\,(\phi - \phi_{\rm att}), \qquad U = \frac{u}{u_{\rm att}(n)}, \qquad H_h = \frac{h}{h_{\rm att}(n)}, \tag{14}$$

where $\phi_{\rm att}(n) = \phi_i - 2n/(cp)$ and $u_{\rm att}, h_{\rm att} \propto e^{n/p}$ are the attractor slices. All three are t-invariant relative coordinates, and the flow (7) reduces *exactly* (not to linear order) to the autonomous system

$$\dot X = \frac{2}{p}\Big(\frac{U}{H_h} - 1\Big), \qquad \dot U = \Big(3 - \frac{1}{p}\Big)U + \frac{1-3p}{p}\,\frac{e^X}{H_h}, \qquad \dot H_h = \frac{1}{p}\Big(\frac{U^2}{H_h} - H_h\Big), \tag{15}$$

with fixed point $(X, U, H_h) = (0, 1, 1)$. Linearising there gives the exact spectrum

$$\lambda_0 = 0, \qquad \lambda_1 = -\frac{1-3p}{p} = -\frac{3}{2}(w-1), \qquad \lambda_2 = -\frac{2}{p} = -3(w+1), \tag{16}$$

all real; the numerical Jacobian reproduces (16) to eight digits for every p tested (Fig. 5). Three readings:

- $\lambda_0 = 0$ with eigenvector $(2, 1, 1)$: the time-translation direction, i.e. the marginal mode of Sec. 5. In these coordinates a time shift is exactly a constant offset—the only memory a converged solution retains.
- $\lambda_1 = -\frac{3}{2}(w-1)$: deviations decay as $a^{3(w-1)/2}$—*exactly half the smoothing exponent* (1). Measured on the full five-variable system for $p = 0.1$ with a 1% velocity perturbation: decay rate 6.99–7.00 per e-fold against the theoretical 7.00, identical in all three coordinates. For $w \gg 1$ a fraction of one e-fold erases generic homogeneous perturbations.
- $\lambda_1 = 0 \iff w = 1$: the scaling solution is linearly *unstable* precisely when $w < 1$, where deviations grow as $a^{3(w-1)/2} \to \infty$ during contraction. The $w = 1$ threshold thus plays a double role: it is both the smoothing condition (1) and the exact stability boundary of the dynamics that performs the smoothing. The $\lambda_2 = -3(w+1)$ mode decays faster still and is numerically invisible after $\sim$0.05 e-folds.

4.2 The Zeno cascade

The recollapse that ends the expanding branch in Sec. 3.3 is itself degenerate: $\rho \to 0$ requires $\frac{1}{2}u^2 = |V|$ to high precision, and at the first recollapse the field velocity has already grown to

Table 3: WKB classicality: fitted exponent of $\mathcal{Q}(a)$ vs the attractor prediction $(1-3p)/p$; $\mathcal{Q}$ at the initial slice and at the GR trust cutoff ($\hbar = 1$).

p	fit / theory	$\mathcal{Q}(t_i)$	$\mathcal{Q}(\rho_{\rm stop})$
0.01	96.99997 / 97	5.0×10^{7}	2.1×10^{-5}
0.05	16.99999 / 17	1.0×10^{7}	5.6×10^{-4}
0.10	6.99999 / 7	5.0×10^{6}	2.9×10^{-2}
0.30	0.33334 / 0.33333	1.7×10^{6}	1.2×10^{5}
0.50	-0.9972 / -1	1.0×10^{6}	5.1×10^{2}

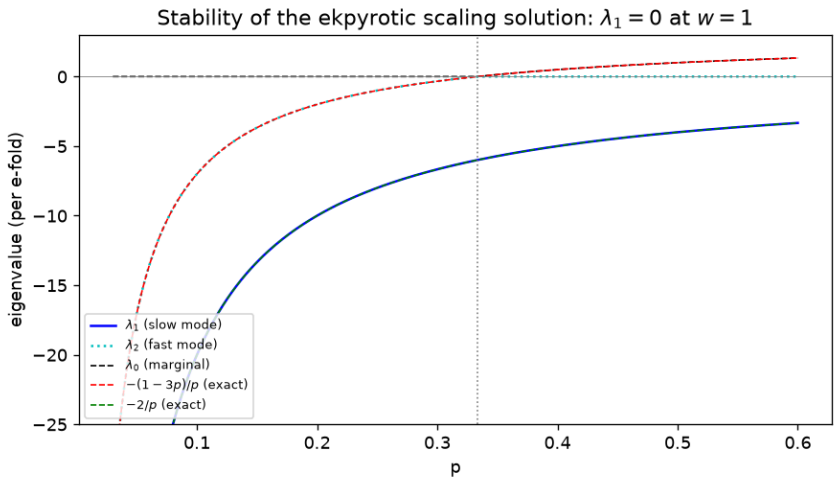


Figure 5: Stability of the ekpyrotic scaling solution. The two nonzero eigenvalues of (16) against p: numerical Jacobian (solid) and exact formulas (dashed) coincide. The slow mode λ_1 crosses zero exactly at $p = 1/3$ ($w = 1$): for $p > 1/3$ the solution that would be doing the smoothing is itself unstable.

$u \approx -711$ against $|V| \approx 2.5 \times 10^5$ (our parameter values)—a mutual cancellation maintained to ten digits by the constraint. Continuing the integration past it (with a floor event at $\rho = 10^{-8}\rho_c$), the effective dynamics generates a sequence of bounce cycles whose durations follow

$$\Delta t_n = \frac{A}{n^2}, \qquad A = \frac{3p}{\pi\sqrt{3\rho_c}}, \qquad |u|_n = \alpha n + \mathcal{O}(1), \quad \alpha = c\pi\sqrt{\tfrac{\rho_c}{3}}, \tag{17}$$

with the field gliding down the potential as $\phi_n = \phi_0 - \frac{2}{c}\ln n$. This law is not merely a fit: it is derivable in three steps, asymptotically in n (where $|V| \gg \rho_c$ at the turnarounds and R is negligible).

(i) Cycle time. Along the effective flow $\rho + p = u^2$ and $H = \sqrt{\rho(1-\rho/\rho_c)/3}$, so the time for one traversal $\rho_c \to 0$ (or $0 \to \rho_c$) at approximately constant u is

$$T_{1/2} = \int_0^{\rho_c} \frac{\mathrm{d}\rho}{3Hu^2} = \frac{1}{\sqrt{3}\,u^2}\int_0^{\rho_c} \frac{\mathrm{d}\rho}{\sqrt{\rho(1-\rho/\rho_c)}} = \frac{\pi\sqrt{\rho_c/3}}{u^2}, \tag{18}$$

the ρ-integral being elementary ($\rho = \rho_c \sin^2\theta$); a full bounce-to-bounce cycle doubles this.

(ii) Velocity growth. Over a cycle, $\Delta u = \oint \dot{u}\,\mathrm{d}t$ with $\dot{u} = -3Hu - c|V|$. The friction term integrates to $-3\oint H\,\mathrm{d}t = -3\,\Delta\ln a \simeq 0$ (a returns to its bounce value), while $|V|$ is nearly constant across a cycle (the field moves by only $\Delta\phi \sim 1/u$). Using the turnaround constraint $|V| = u^2/2 - \rho_c + R \simeq u^2/2$, the u^2 *cancels*:

$$\Delta u \simeq -c\,|V|\,T = -c\,\frac{u^2}{2}\,\frac{2\pi\sqrt{\rho_c/3}}{u^2} = -c\pi\sqrt{\tfrac{\rho_c}{3}} \;\equiv\; -\alpha. \tag{19}$$

(iii) The law. Substituting $u_n = \alpha n$ into the doubled (18) gives (17) with $A = 2\pi\sqrt{\rho_c/3}/\alpha^2 = 3p/\pi\sqrt{3\rho_c}$: the exponent $\gamma = 2$ is *exact*, and A carries no free parameters. The field drift follows as $\Delta\phi_n = \oint u\,\mathrm{d}t \simeq -u_n T_n = -2/(cn)$, i.e. a harmonic (not geometric) creep, and the accumulation tail is $t_{\mathrm{acc}} - t_N = A\,(\pi^2/6 - H_N^{(2)}) \simeq A/N$.

All four predictions are confirmed numerically (Fig. 8 and Fig. 9): the growth rate α to four significant digits for both $p = 0.1$ (5.1938 vs $c\pi\sqrt{\rho_c/3} = 5.1939$) and $p = 0.05$ (7.3452 vs 7.3453); the scaled cycle time $\Delta t_n u_n^2/2\pi\sqrt{\rho_c/3} = 1.025 \pm 0.022$ and 1.026 ± 0.022 (the excess being the $1/n$ transient at small n); the drift law $\Delta\phi_n/-2/(cn) = 1.004$ and 1.002;

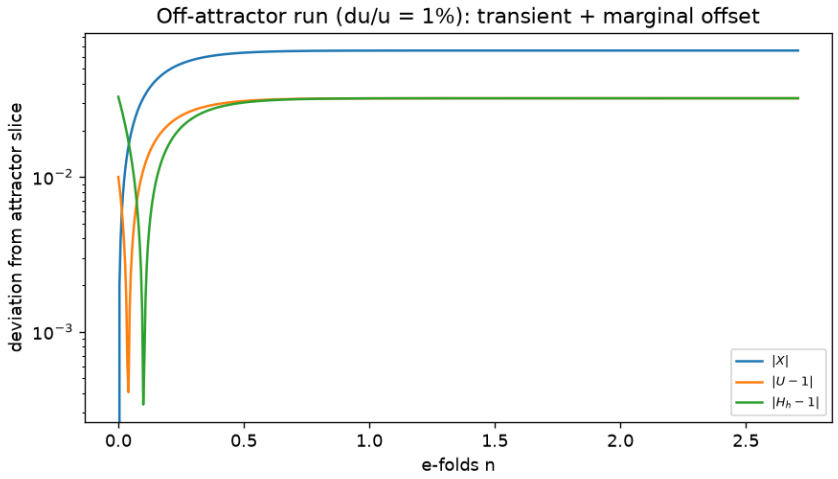


Figure 6: Convergence onto a shifted attractor. Deviations of the three scaled coordinates from the attractor slice, for the full system with a 1% velocity perturbation ($p = 0.1$). Each curve is a decaying transient (e^{-7n}) on top of a constant offset—the marginal component, which is a pure time shift.

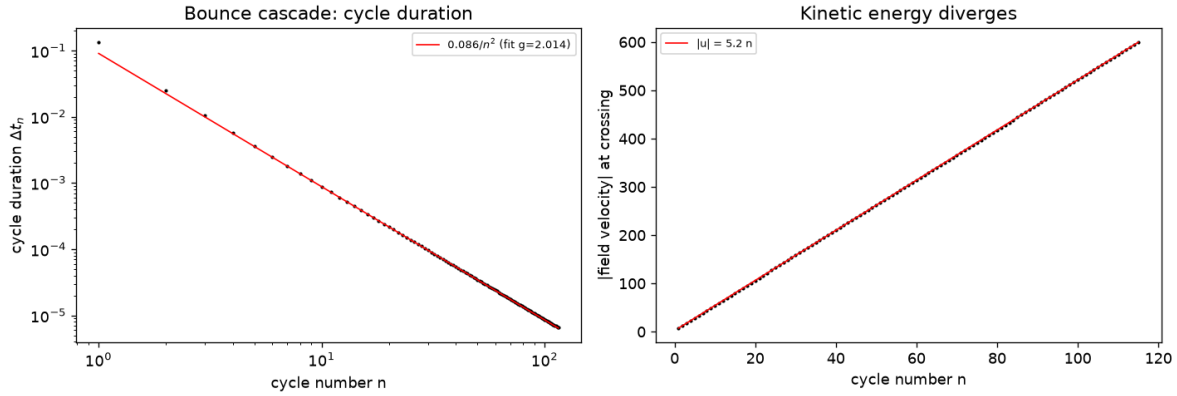


Figure 7: The bounce cascade. Left: cycle durations Δt_n against cycle number, with the fitted law $\Delta t_n = 0.086/n^{2.01}$. Right: field velocity at the crossings, $|u|_n = 5.2\,n+2$. The $\sum n^{-2}$ series accumulates at $t_{\rm acc} \approx -3.6254$: infinitely many bounces with unbounded kinetic energy in finite time.

and the accumulation time from the analytic tail sum, $t_{\rm acc} = -3.625382$, matching the earlier series estimate. The $A \propto p$ scaling is also confirmed (0.0443 measured vs 0.0431 predicted at $p = 0.05$).

We emphasise that densities never leave $[0, \rho_c]$ in this sequence; it is the velocity that diverges, which is why the breakdown is easy to miss if one monitors only ρ. The cascade is thus a Zeno-type pathology of the closure (8) glued to a negative potential, now with a quantitatively explained structure.

4.3 A failed remedy: flipping the sign of the potential

A natural guess is that the recollapse is simply the fault of $V < 0$, and that letting the potential turn positive at the bounce (a crude stand-in for reheating) would sustain the expansion. It does not. At the bounce the field velocity is inherited from the contraction, $u_b = -2.44$, pointing toward larger $|V|$; with $V > 0$ in the expanding branch the field equation keeps driving ϕ toward more negative values, $|V| = V_0 e^{-c\phi}$ grows, and the on-shell solution is driven back across the constraint boundary $\rho = \rho_c$ within $\Delta t \lesssim 0.2$ of the bounce, where the integration

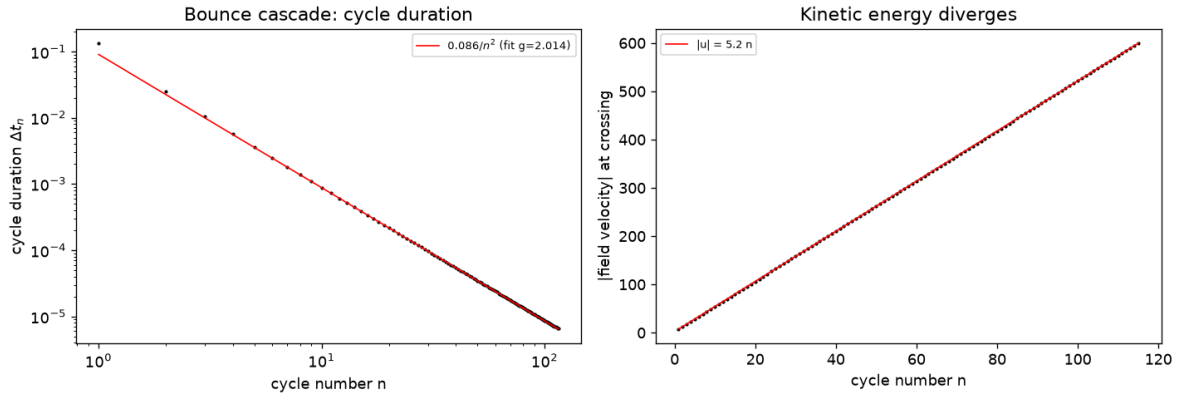


Figure 8: The bounce cascade. Left: cycle durations Δt_n against cycle number, with the fitted law $\Delta t_n = 0.086/n^{2.01}$. Right: field velocity at the crossings, $|u|_n = 5.2\,n+2$. The $\sum n^{-2}$ series accumulates at $t_{\rm acc} \approx -3.6254$: infinitely many bounces with unbounded kinetic energy in finite time.

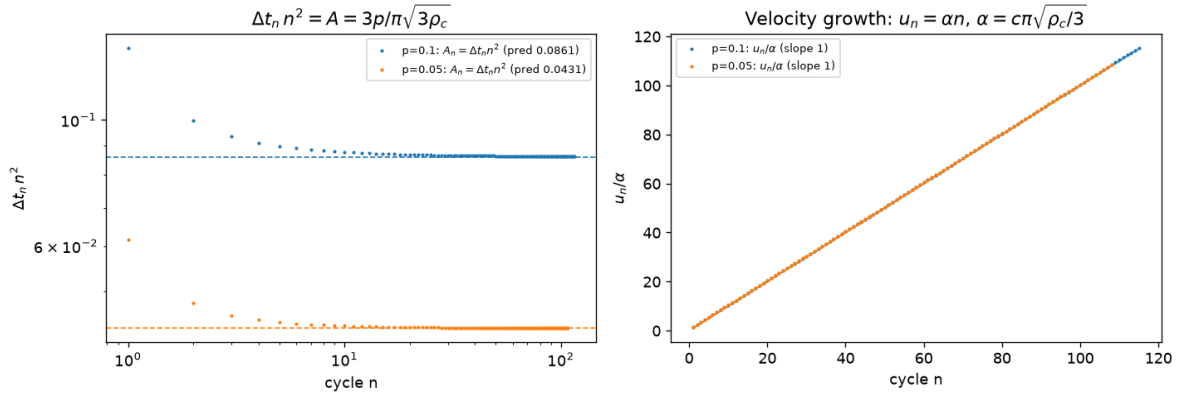


Figure 9: The cascade law, derived vs measured. Left: $\Delta t_n n^2$ converges onto the predicted $A = 3p/\pi\sqrt{3\rho_c}$ (dashed) for both p. Right: u_n/α is a unit-slope line, $\alpha = c\pi\sqrt{\rho_c/3}$, for both p.

breaks down. Between the Zeno cascade ($V < 0$) and this runaway ($V > 0$), the conclusion is that *no* continuation of (8) with an unchanged scalar sector produces a valid expanding phase: a working bounce must alter the field's post-bounce roll—as the ghost-condensate mechanism of Ref. [3] (asymptotic approach to the top of the potential) and galileon constructions [4] (modified kinetic sector) in fact do.

5 A marginal mode of the ekpyrotic attractor

Our accuracy study exposed a conditioning property of the scaling solution that matters for any numerical treatment of ekpyrotic backgrounds, independent of the physics conclusions above. Its linear origin is the zero eigenvalue of the spectrum (16): the time-translation direction, which is exactly marginal in the t-invariant variables.

The family (3) has an exact time-translation symmetry: given any solution, $t \to t + \delta t_c$ is another. Linearising about the attractor, a homogeneous perturbation decomposes into stable directions (decaying relative to the background as powers of a) plus the symmetry direction, whose effect is to shift the time coordinate. In e-fold time the background time behaves as $t_c - |t| = |t_c - t_i|e^{-n/p}$, so an initial time offset appears as

$$\delta t(n) = \delta t_{c,0}\, e^{n/p}, \tag{20}$$

i.e. the time-translation component is *exactly marginal* in the physical (t-invariant) variables—it neither grows nor decays relative to the background—but appears as an exponentially growing offset in the t coordinate itself. Consequently, *any* homogeneous perturbation of the attractor—an offset of the initial data, or the small change in $\dot{H}$ from the shear term—acquires a growing error in the measured crunch time, while convergence to the attractor in all physical variables is unaffected.

We verified this quantitatively by integrating the GR system in cosmic time from attractor initial data with H_i on the constraint surface and shear fraction $f \in \{0, 10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}\}$: the displacement of the measured crunch time is exactly linear,

$$\frac{\delta t_c}{|t_i|} \simeq -36.8\, f \qquad (p = 0.1), \tag{21}$$

over three decades of f, and independent of the integrator tolerance down to rtol 10^{-13}—it is the physical response of the homogeneous system to shear backreaction, not solver noise. For $p = 0.1$ and $f = 10^{-4}$ the crunch sits at $t_c \simeq -3.95$ instead of the unperturbed -0.27.

Two practical corollaries follow. First, comparisons between contracting solutions (e.g. GR versus modified dynamics, as in Sec. 3.3) must be made in crunch-centred time $t - t_c$ or, better, in t-invariant variables (a, n, ρ); in absolute t, the marginal mode masquerades as a genuine dynamical discrepancy. Second, cosmic-time integration is intrinsically fragile deep in the contraction: relative errors at the working-precision level excite the mode, and no tolerance recovers the target t_c; e-fold time removes the issue for all quantities except t itself, and t never carries physical information in this problem.

6 Assumptions, predictions, and what is left open

Assumed without derivation. (i) flat slicing and a single canonical scalar with an exactly exponential potential; (ii) homogeneous, adiabatically evolved shear (exact for Bianchi I; inhomogeneous anisotropies, with their growing a^{-2}-gradient modes [2], are excluded); (iii) the effective Friedmann equation (8), specified in the style of LQC effective dynamics with continuity unchanged, and no claim of derivation; (iv) the first-derivative-free WDW factor ordering and the minisuperspace truncation; (v) initial data exactly on the scaling attractor. Item (v) is mild—the attractor is known analytically and our runs spot-check its stability—but a systematic off-attractor initial-data sweep is listed below as open.

What the models predict, as verified here. (1) Smoothing exponent $3(w-1)$ exactly; suppression linear in the initial shear fraction, insensitive to V_0, completed within $\mathcal{O}(1)$ e-folds for $w \gg 1$. Generic homogeneous perturbations decay as $a^{3(w-1)/2}$, retaining only a time shift; the scaling solution is linearly stable iff $w > 1$. (2) No GR bounce; a specified H^2-deformed dynamics bounces exactly at $\rho = \rho_c$ and inherits the smoothed state, but recollapses immediately for $V < 0$ ($a_{\max}/a_b = 1.0115$); its continuation is a Zeno cascade ($\Delta t_n \propto n^{-2}$, $|u|_n \propto n$) accumulating in finite time, and flipping the sign of the potential does not avert breakdown. (3) WKB classicality exponent $(1-3p)/p$ exactly—equivalently $e^{-(\epsilon-3)\mathcal{N}/(\epsilon-1)}$—so the contracting branch becomes arbitrarily classical, in the WKB sense, as the density grows, for all $w > 1$, and progressively fails to for $w < 1$.

What remains unverified or open. (i) A viable post-bounce history: matching to a hot expanding phase requires physics absent here (ghost-condensate/galileon-type bounce mechanics, reheating, or a potential that turns positive); our effective model demonstrates the failure mode explicitly rather than resolving it, and Sec. 4.3 shows the failure is robust to a sign flip of the potential. (ii) The cascade law $\Delta t_n \propto n^{-2}$ of Sec. 4.2 is now derived analytically and verified to four significant digits, including its p-scaling; what remains open there is only the $1/n$

correction terms (the few-percent transient at small n). (iii) Perturbations across the bounce and the quantum boundary proposal (no-boundary vs tunnelling, cf. [7, 8]): our Sec. 3.4 is a trajectory-level WKB check, the input to—not a substitute for—the saddle-point analyses. (iv) Inhomogeneous anisotropy gradients. (v) Off-attractor initial data in the anisotropic sector: Sec. 4.1 settles the pure-scalar case (decay at $3(w-1)/2$ plus a time shift); the corresponding statement with mixmaster-type shear excitations is open.

7 Conclusions

Within its stated assumptions, the classical ekpyrotic mechanism is exactly what it claims to be: contraction with $w > 1$ drives the universe to an isotropic, scalar-dominated attractor, with the anisotropy ratio suppressed as $a^{3(w-1)}$, to numerical precision far beyond what is physically relevant, and within a handful of contraction e-folds. The dynamics is even better behaved than the smoothing law alone suggests: in t-invariant relative coordinates the flow about the attractor has the exact spectrum $\{0, -\frac{3}{2}(w-1), -3(w+1)\}$, so generic homogeneous perturbations die as $a^{3(w-1)/2}$ within a fraction of an e-fold, and the only memory they leave is a time shift—while for $w < 1$ the same spectrum turns unstable, the $w = 1$ threshold thus governing both smoothing and stability. Neither the contraction nor the smoothing contains an exit: reading a bounce out of the background requires wholesale new dynamics, and the naive option—an H^2-deformation glued onto the unchanged negative-potential model—works exactly once and then visibly falls apart, recollapsing into a Zeno cascade whose law we derive and confirm ($\Delta t_n = 3p\, n^{-2}/\pi\sqrt{3\rho_c}$, $|u|_n = c\pi\sqrt{\rho_c/3}\, n$: infinite bounces with unbounded kinetic energy in finite time) that no sign flip of the potential averts. Semiclassically, the contracting branch is an increasingly good WKB history: $\mathcal{Q} \propto a^{(1-3p)/p}$ improves monotonically through ekpyrosis, exponentially in the contraction e-folds, providing the local, trajectory-level confirmation of the classicality behaviour found in the no-boundary analyses of Ref. [6]. Finally, we have quantified a numerical property of the ekpyrotic attractor with practical consequences beyond this study: its marginal time-translation mode amplifies any homogeneous perturbation into a strictly linear crunch-time displacement, $\delta t_c \propto f$, which dictates the choice of integration variables and warns against absolute-time comparisons of contracting solutions.

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